

Ptbp1 is not required for retinal neurogenesis and cell fate specification

Reviewed Preprint

v1 • September 19, 2025

Not revised

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eLife Assessment

This study used a conditional knockout mouse line to remove *Ptbp1* in retinal progenitors and showed that its deletion has no effect on retinal neurogenesis or cell fate specification, thereby challenging the prevailing view of *Ptbp1* as a master regulator of neuronal fate. The findings are **convincing**, supported by transcriptome analysis, histology, and proliferation assays. This study is **important**, though the genetic tools employed may not fully capture *Ptbp1*'s potential role during the earliest stages of retinal development.

<https://doi.org/10.7554/eLife.108331.1.sa2>

Abstract

The RNA-binding protein *Ptbp1* has been proposed as a master regulator of neuronal fate, repressing neurogenesis through its effects on alternative splicing and miRNA maturation. While prior studies using RNA interference suggested that *Ptbp1* loss promotes neurogenesis, recent genetic studies have failed to replicate glia-to-neuron conversion following *Ptbp1* loss of function. To evaluate the role of *Ptbp1* in developmental neurogenesis *in vivo*, we conditionally disrupted *Ptbp1* in retinal progenitors. *Ptbp1* was robustly expressed in both retinal progenitors and Müller glia but absent from postmitotic neurons, and efficient loss of function in mutant animals was confirmed using immunostaining for *Ptbp1*. Furthermore, bulk RNA-Seq at E16 revealed accelerated expression of late-stage progenitor and photoreceptor-specific genes and altered splicing patterns in *Ptbp1* mutants, including increased inclusion of rod photoreceptor-specific exons. However, we observed no defects in retinal lamination, progenitor proliferation, or cell fate specification in mature retina. ScRNA-Seq of mature mutant retinas revealed only modest transcriptional changes which partially

recapitulate alterations seen following selective deletion of *Ptbp1* in mature glia. Our findings demonstrate that *Ptbp1* is dispensable for retinal development and suggest that its proposed role as a central repressor of neurogenesis should be reevaluated.

Introduction

Control of neurogenesis in progenitor cells of the developing CNS involves an elaborate interplay between negative and positive regulators. Neurogenic bHLH factors such as *Ascl1* and *Neurog2* induce neural progenitors to undergo terminal neurogenic divisions, while inhibitory factors such as components of the Notch pathway and NFI family transcription factors promote proliferative divisions and gliogenesis (Hufnagel et al. 2013 [↗](#); Keeley et al. 2023 [↗](#); Yaron et al. 2006 [↗](#); Clark et al. 2019 [↗](#); Lyu et al. 2021 [↗](#)). Studies in species such as zebrafish that undergo injury-induced neurogenesis by reprogramming of endogenous glial cells find that this process requires both upregulation of neurogenic factors and downregulation of inhibitory factors (Fausett, Gumerson, and Goldman 2008 [↗](#); Briona et al. 2015 [↗](#); Lyu et al. 2023 [↗](#)). This has recently inspired many studies aimed at inducing neurogenic competence in mammalian glia by selectively targeting these factors (Jorstad et al. 2017 [↗](#); Hoang et al. 2020 [↗](#); Todd et al. 2022 [↗](#), 2021 [↗](#); Le et al. 2025 [↗](#), 2024 [↗](#)).

One factor that has attracted considerable interest is the RNA binding protein and splicing regulator *Ptbp1*, which is widely expressed in non-neuronal cells, including both neural progenitors and glia (Lilleväli, Kulla, and Ord 2001 [↗](#); Boutz et al. 2007 [↗](#)). It has been reported that *Ptbp1* acts as a broad inhibitor of neuronal-specific splicing (Boutz et al. 2007 [↗](#); Ling et al. 2016 [↗](#)), and inhibits maturation of miR-124a, which strongly promotes neurogenesis (Makeyev et al. 2007 [↗](#)), while its paralog *Ptbp2* promotes neuronal-specific splicing (Licatalosi et al. 2012 [↗](#); Li et al. 2014 [↗](#)). It was also proposed that *Ptbp1* acts as a master regulator of developmental neurogenesis in the CNS (Shibasaki et al. 2013 [↗](#)). This then inspired studies which used *Ptbp1* knockdown to attempt to reprogram glia into functional neurons. While it was reported that *Ptbp1* knockdown *in vivo* induced robust generation of functional retinal ganglion cells from Müller glia (Zhou et al. 2020 [↗](#)), as well as functional dopaminergic neurons from midbrain astrocytes (Qian et al. 2020 [↗](#)), attempts to replicate these results by other groups were unsuccessful (L.-L. Wang et al. 2021 [↗](#); Hoang et al. 2023 [↗](#), 2022 [↗](#); Chen et al. 2022 [↗](#); Xie, Zhou, and Chen 2022 [↗](#)). Notably, genetic loss of function of *Ptbp1* coupled with scRNA-Seq analysis failed to observe either loss of expression of glial-specific genes or induction of neuron-specific genes in mutant glia (Hoang et al. 2023 [↗](#), 2022 [↗](#)).

This result raises questions about previous claims about the central role of *Ptbp1* in controlling developmental neurogenesis. Previous functional studies of the role of *Ptbp1* in neuronal development were primarily conducted *in vitro* using RNAi to interrogate *Ptbp1* function (Spellman, Llorian, and Smith 2007 [↗](#); Zheng et al. 2012 [↗](#); Linares et al. 2015 [↗](#)). The few *in vivo* studies of *Ptbp1* function in developing CNS have also almost exclusively used RNAi, and generally report modest and inconsistent phenotypes (X. Zhang et al. 2016 [↗](#)). Conventional *Ptbp1* homozygous mutants are lethal, precluding direct analysis of its role in developmental neurogenesis (Shibayama et al. 2009 [↗](#)). While one study reported accelerated cortical neurogenesis in *Ptbp1* heterozygotes (Shibasaki et al. 2013 [↗](#)), these effects were also quite modest and difficult to interpret, given the broader role of *Ptbp1* in early embryonic development.

The retina is a highly accessible and tractable model for the study of CNS neurogenesis and cell fate specification (Ptito, Bleau, and Bouskila 2021 [↗](#)), and conditional mutagenesis allows selective loss of function of any gene of interest in retinal progenitors without affecting viability (Sauer 1998 [↗](#)). We sought to use this approach to investigate the role of *Ptbp1* in regulating neurogenesis and cell fate specification in developing retina. While we observe changes in RNA splicing and a moderate acceleration of retinal development as measured by bulk RNA-Seq, we do not observe

any changes in proliferation or neurogenesis during retinal development, or changes in the cell type composition of *Ptbp1*-deficient retinas. Furthermore, we observe only modest changes in gene expression in adult mutant retina using scRNA-Seq analysis. These results demonstrate that *Ptbp1* is fully dispensable for retinal development.

Results

***Ptbp1* mRNA is primarily expressed in retinal progenitors and Müller glia**

To examine the expression patterns of *Ptbp1* and *Ptbp2* in developing and adult neuroretina, we analyzed scRNA-Seq datasets from mouse and human retina that had been previously generated by our lab (Clark et al. 2019 [↗](#); Lu et al. 2020 [↗](#)). We observe robust expression of *Ptbp1* mRNA in primary retinal progenitors and Müller glia of both species, with weaker expression in neurogenic progenitors, and little expression detectable in neurons at any developmental age. In contrast, *Ptbp2* is expressed in both progenitors and neurons, although with somewhat higher expression in retinal ganglion and amacrine cells (Fig. S1a [↗](#)). We observe that *Ptbp1* is expressed in primary progenitors at all developmental ages between E12 and P8 in mice and 9 and 20 gestational weeks in humans, although expression levels are moderately increased at later stages of neurogenesis (Fig. S1b [↗](#)).

***Chx10-Cre;Ptbp1^{lox/lox}* mice show loss of *Ptbp1* immunoreactivity in developing and mature retina**

To selectively disrupt *Ptbp1* function in retinal progenitors, we generated *Chx10-Cre;Ptbp1^{lox/lox}* mice. In these mice, Cre recombinase is expressed broadly in retinal progenitors from the beginning of neurogenesis onwards (Rowan and Cepko 2004 [↗](#)), allowing efficient deletion of the conditional mutant allele of *Ptbp1* (Shibasaki et al. 2013 [↗](#)) (Fig. 1a [↗](#)). Mutant mice were born at expected Mendelian ratios and showed no gross defects in retinal morphology. To confirm efficient deletion of *Ptbp1* in retinal progenitors, we performed immunohistochemistry for *Ptbp1* at E14, P1, and P30 in both control *Chx10-Cre;Ptbp1^{+/+}* and homozygous mutant *Chx10-Cre;Ptbp1^{lox/lox}* mice (Fig. 1b-d [↗](#)). At all stages, we observed a loss of *Ptbp1* protein expression in retinal progenitors and Müller glia, with a more than ten-fold reduction in the number of immunoreactive Müller glia observed at P30 (Fig. 1e [↗](#)). At both E14 and P1, we also observed a variable level of immunoreactivity in cells in the ganglion cell layer in control mice, which was also seen in *Chx10-Cre;Ptbp1^{lox/lox}* mice, suggesting that this may represent crossreactivity with an unknown epitope. This ganglion cell layer staining was not observed at P30, however. This demonstrates that *Ptbp1* is efficiently and selectively deleted in retinal progenitors from early stages of neurogenesis.

Embryonic *Chx10-Cre;Ptbp1^{lox/lox}* mice show shifts in temporal identity, increased expression of genes specific to photoreceptors, and altered patterns of RNA splicing

To further investigate the effects of *Ptbp1* deletion on embryonic progenitors, we conducted bulk RNA-Seq on whole retina from E16 wildtype, *Chx10-Cre;Ptbp1^{lox/+}*, and *Chx10-Cre;Ptbp1^{lox/lox}* retinas. A comparison of heterozygous *Chx10-Cre;Ptbp1^{lox/+}* and homozygous *Chx10-Cre;Ptbp1^{lox/lox}* mutants identified 2075 genes that were significantly differentially expressed (Fig. 2a [↗](#), Extended Data 1). As expected, homozygotes showed a loss of *Ptbp1* expression along with a compensatory upregulation of *Ptbp2*, as has previously been observed following selective disruption of *Ptbp1* in mature Müller glia (Hoang et al. 2022 [↗](#)) (Fig. 2b [↗](#)). In addition, we observed a modest increase in expression of multiple genes specific to both rods and cone photoreceptors – including *Opn1sw*

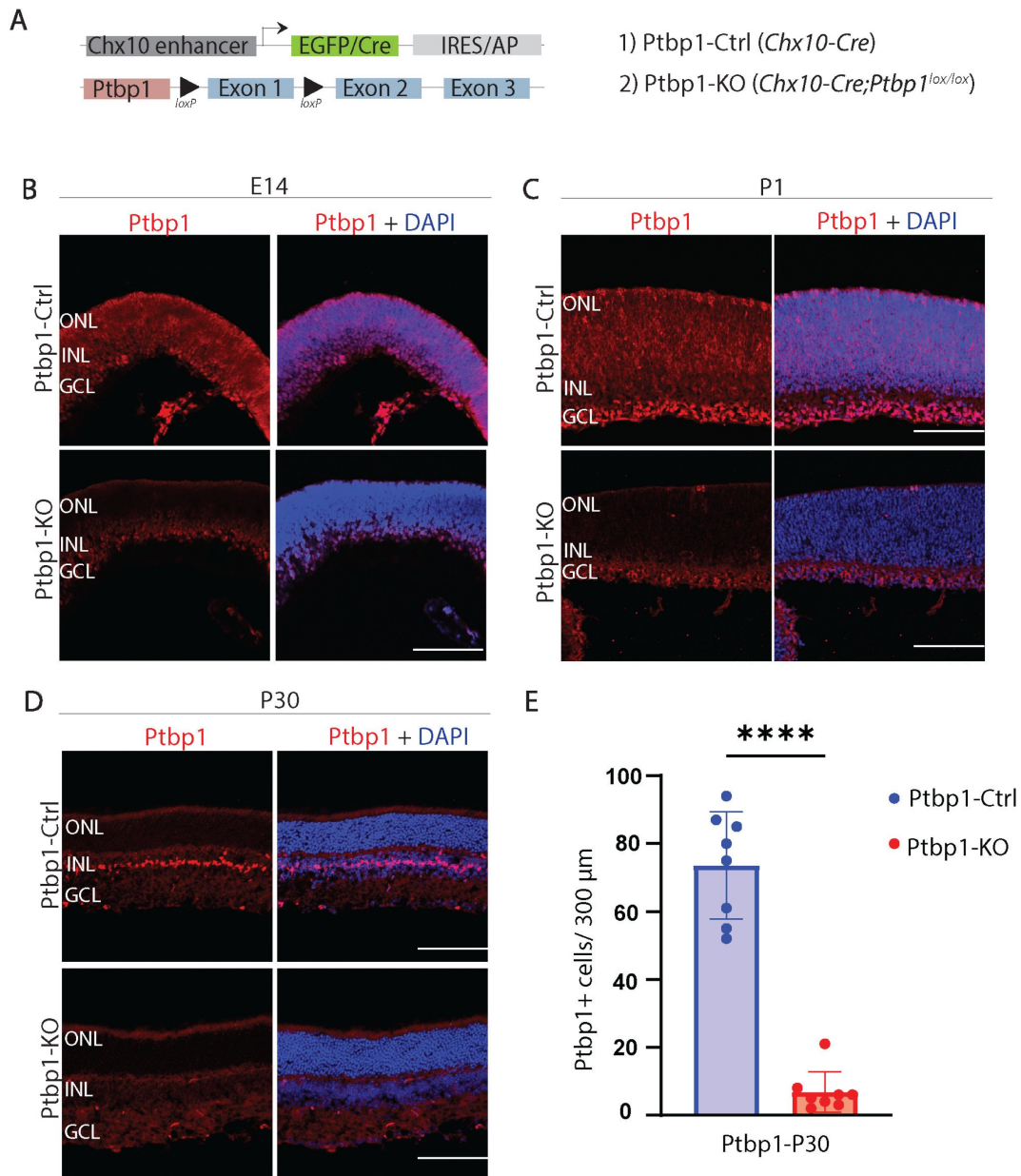


Figure 1.

***Ptbp1* is successfully deleted in *Chx10-Cre;Ptbp1^{lox/lox}* mice in developing and mature retina.**

(A) A schematic diagram of the generation of specific deletion of *Ptbp1* in early RPCs using *Chx10-Cre*. (B) Representative immunostaining for *Ptbp1* expression in *Chx10-Cre;Ptbp1^{+/+}* (Ptbp1-Ctrl) and *Chx10-Cre;Ptbp1^{lox/lox}* (Ptbp1-KO) mice at E14.5, (C) P1, (D) and P30. (E) Quantification of Ptbp1-positive cells in Ptbp1-Ctrl and Ptbp1-KO retinas (n=8/genotype). Significance was determined via unpaired *t* test: **** *p*<0.0001. Each data point was calculated from an individual retina. ONL, outer nuclear layer; INL, inner nuclear layer; GCL, ganglion cell layer. Scale bar= 100 μm.

, and *Chx10-Cre;Ptbp1^{lox/lox}* retinas. A comparison of heterozygous *Chx10-Cre;Ptbp1^{lox/+}* and homozygous *Chx10-Cre;Ptbp1^{lox/lox}* mutants identified 2075 genes that were significantly differentially expressed (Fig. 2a, Extended Data 1). As expected, homozygotes showed a loss of *Ptbp1* expression along with a compensatory upregulation of *Ptbp2*, as has previously been observed following selective disruption of *Ptbp1* in mature Müller glia (Hoang et al. 2022) (Fig. 2b). In addition, we observed a modest increase in expression of multiple genes specific to both rods and cone photoreceptors – including *Opn1sw* and *Rcvrn* – as well as altered expression of multiple genes selectively expressed in either early or late-stage RPCs (Clark et al. 2019; Lu et al. 2020; Lyu et al. 2021). For instance, the early-stage RPC-specific genes *Sfrp2* and *Foxp1* were downregulated, while late-stage RPC-enriched genes such as *Sox8*, *Nfix*, and *Hes5* were upregulated (Fig. 2b). Gene Ontology (GO) enrichment analysis likewise showed significant upregulation of genes involved in photoreceptor cell development and genes controlling synaptic function (Fig. 2c). However, immunohistochemical analysis does not detect any clear changes in the expression of HuC/D+NeuN, Otx2 or *Opn1sw* in *Chx10-Cre;Ptbp1^{lox/lox}* homozygous mutants relative to *Chx10-Cre;Ptbp1^{+/+}* control at either E14 or P0, and the number of Otx2-positive cells at P0 was not altered (Fig. S2). This implies that while at the transcriptional level retinal development in embryonic *Ptbp1* mutants appears to be moderately accelerated, based on higher expression of both markers of late-stage retinal progenitors and photoreceptors, this does not result in an increase in number of photoreceptors at this stage.

In light of the established role of *Ptbp1* in regulating RNA splicing in non-neuronal cells, we analyzed splicing differences in *Ptbp1*-deficient retinas using our dataset. We identified 864 differential percent-spliced-in (PSI) events, including changes in retained introns, alternative 5' or 3' splice site usage, mutually exclusive exons, and exon skipping—consistent with the known role of *Ptbp1* as a splicing repressor (Fig. 2d–e, Extended Data 1) (Vuong et al. 2016; Yap et al. 2012). To assess whether these splicing changes reflect neuronal or rod photoreceptor maturation, we compared our differential splicing events to those observed in adult neurons and rods (Ling et al. 2020). We found that 8 splicing events (7%) known to be enriched in central nervous system neurons and 19 events (35%) specific to rods overlapped with the splicing changes observed in *Ptbp1*-deficient retinas. (Fig. 2f, Extended Data 1). Notably, splicing patterns in *Ptbp1*-deficient retinas showed stronger correlation with Thy1-positive neurons—which exhibit low *Ptbp1* expression—and minimal overlap with microglia and auditory hair cells, the adult cell types with the highest *Ptbp1* levels (Fig. S3).

***Chx10-Cre;Ptbp1^{lox/lox}* mice show no changes in retinal progenitor proliferation or neurogenesis**

To determine whether retinal progenitor-specific loss of function of *Ptbp1* resulted in a depletion of retinal progenitors, we conducted EdU labeling to quantify the number of cells in S-phase in P0 retina (Fig. 3a). We did not observe any significant differences in the number of EdU-positive cells between wildtype controls and *Chx10-Cre;Ptbp1^{lox/lox}* mice (Fig. 2b,c). To determine whether loss of function of *Ptbp1* influenced cell type specification in developing retina, we conducted immunostaining for a range of cell type-specific markers in control and *Chx10-Cre;Ptbp1^{lox/lox}* mice (Fig. 4). Although *Ptbp1* immunoreactivity was lost in homozygous mutant retinas (Fig. 4a), we did not observe any significant changes in the numbers or distribution of Sox9-positive Müller glia (Fig. 4b,f), HuC/D-positive amacrine and ganglion cells (Fig. 4c,f), Otx2-positive bipolar cells and photoreceptors (Fig. 4c,f), Rbpms-positive retinal ganglion cells (Fig. 4d,f), or cone arrestin-positive cone photoreceptors (Fig. 4e,f). Taken together, these findings show that *Ptbp1* loss of function induces neither precocious cell cycle exit nor affects normal levels and patterns of developmental neurogenesis and cell fate specification.

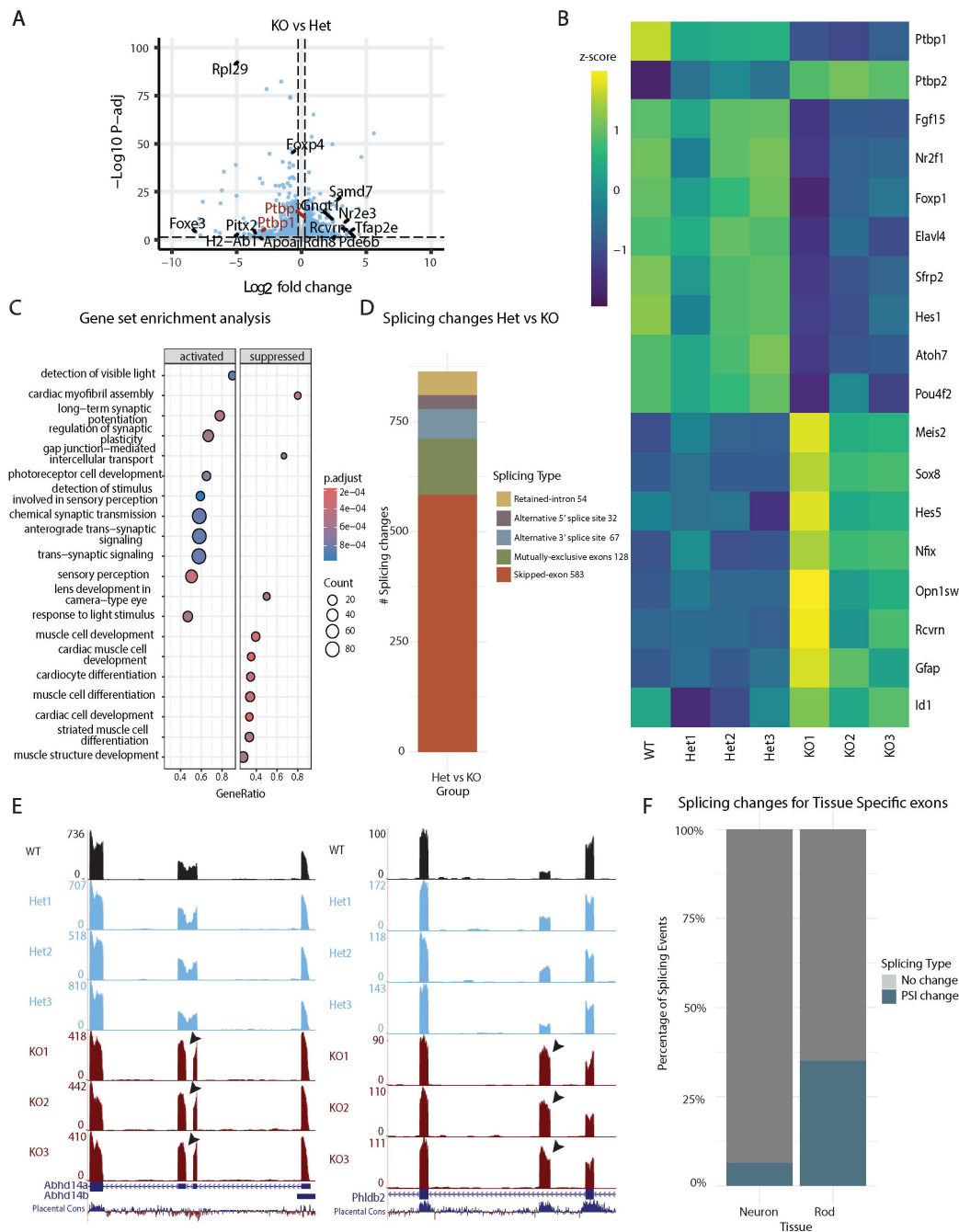


Figure 2.

BulkRNA-seq analysis reveals accelerated differentiation and altered RNA splicing in *Chx10-Cre;Ptbp1^{lox/lox}* mice at E16.5.

(A) Volcano plot showing 2075 differentially expressed genes between heterozygous *Chx10-Cre;Ptbp1^{lox/+}* and homozygous *Chx10-Cre;Ptbp1^{lox/lox}* mutants as measured by bulk RNA-seq. **(B)** Heatmap of select differentially expressed genes in wildtype *Chx10-Cre;Ptbp1^{+/+}*, heterozygous *Chx10-Cre;Ptbp1^{lox/+}*, and homozygous *Chx10-Cre;Ptbp1^{lox/lox}* samples. **(C)** Gene set enrichment analysis showing activated and repressed gene ontology terms in the homozygous *Ptbp1* mutant samples. **(D)** Barplot showing 864 splicing changes in homozygous *Ptbp1* mutant samples by splicing class including retained-introns (54 events), alternative 5' splice site (32 events), alternative 3' splice site (67 events), mutually-exclusive exons (128 events) and skipped-exons (583 events). **(E)** Track plots showing examples of alternative splicing of exons (arrows) in the *Ptbp1* mutant samples. **(F)** Bar plot showing that 7% of neuron-enriched and 35% of rod-specific splicing events overlap with splicing changes observed in *Ptbp1* mutants.

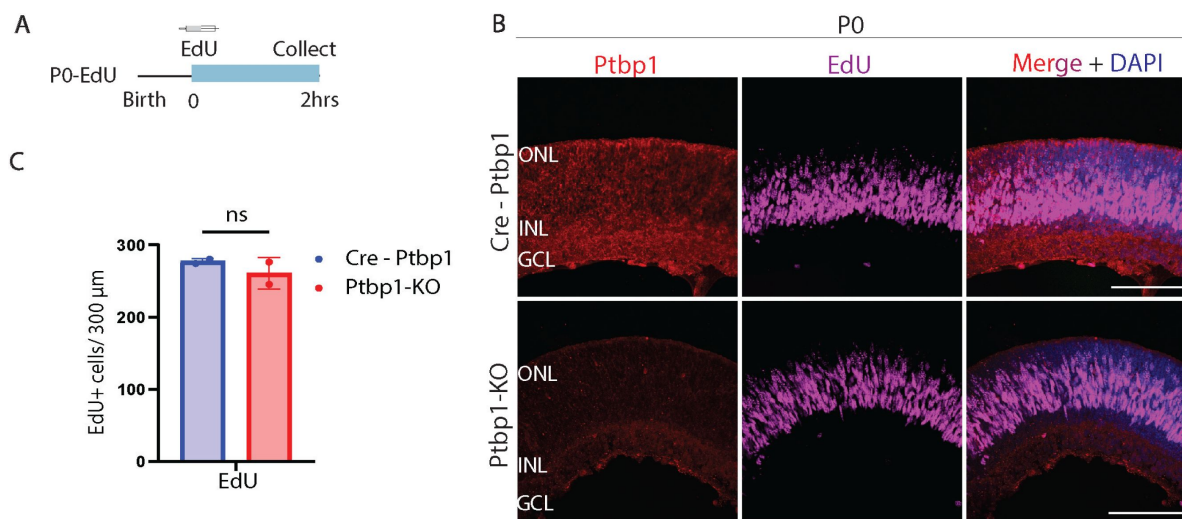


Figure 3.

***Ptbp1* deletion does not alter retinal progenitor proliferation.**

(A) A schematic diagram of the experimental timeline for labeling dividing cells using EdU. **(B)** Representative immunostaining for Ptbp1 expression and EdU labeling in Ptbp1-Ctrl and Ptbp1-KO mice at P0. **(C)** Quantification of EdU-positive cells in Ptbp1-Ctrl and Ptbp1-KO retinas ($n=2/\text{genotype}$). Significance was determined via unpaired t test: $ns=p>0.05$. Each data point was calculated from an individual retina. ONL, outer nuclear layer; INL, inner nuclear layer; GCL, ganglion cell layer. Scale bar=100 μ m.

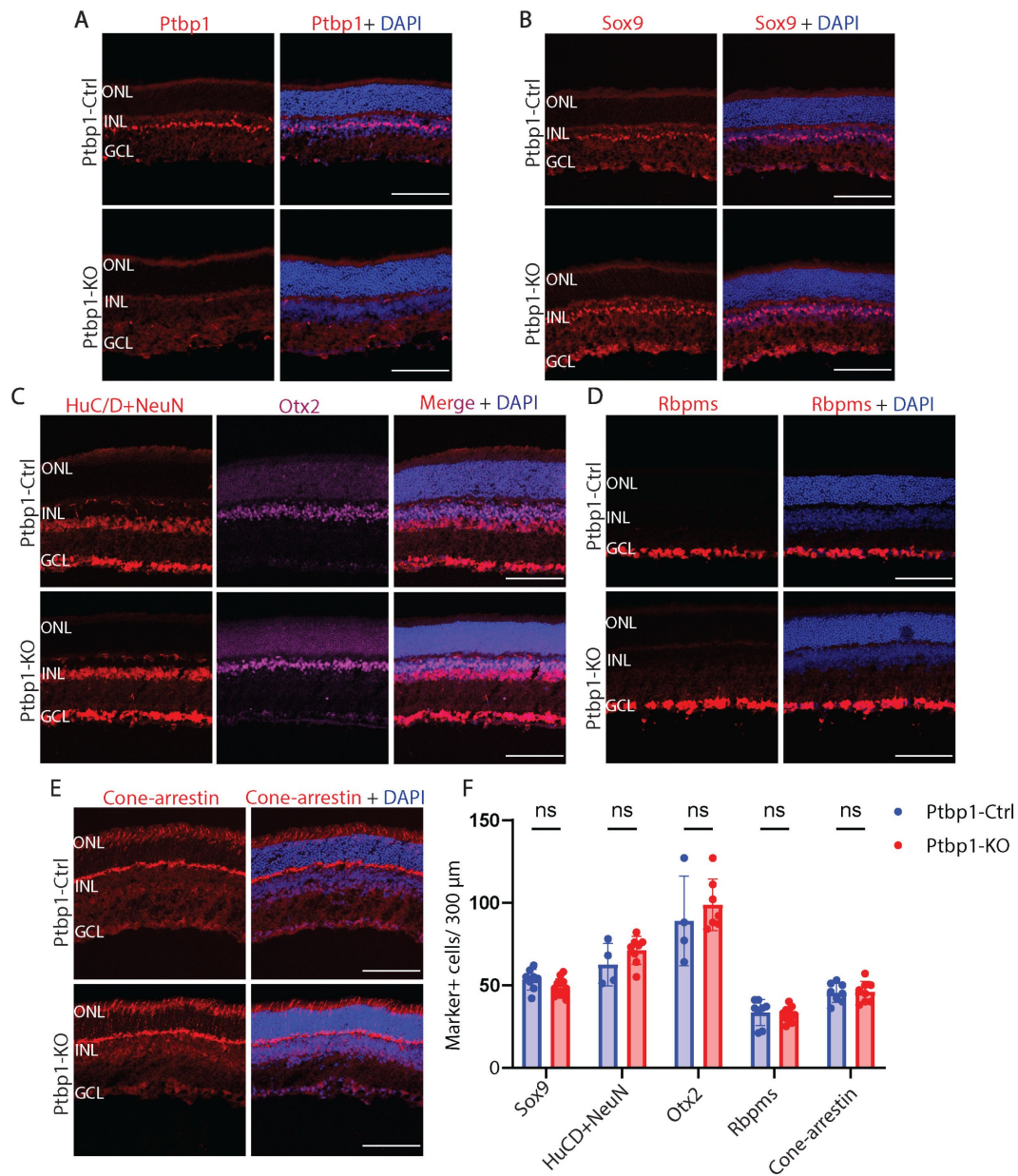


Figure 4.

***Ptbp1* deletion does not alter retinal neurogenesis.**

(A) Representative immunostaining for *Ptbp1*, (B) *Sox9*, (C) *HuC/D+NeuN*, *Otx2*, (D) *Rbpms*, and (E) *Cone-arrestin* in *Ptbp1*-Ctrl and *Ptbp1*-KO mice at P30. (F) Quantification of *Sox9*, *HuC/D+NeuN*, *Otx2*, *Rbpms*, and *Cone-arrestin*-positive cells in *Ptbp1*-Ctrl and *Ptbp1*-KO retinas ($n \geq 4$ /genotype). Significance was determined via multiple unpaired *t* tests: ns= $p > 0.05$. Each data point was calculated from an individual retina. ONL, outer nuclear layer; INL, inner nuclear layer; GCL, ganglion cell layer. Scale bar= 100 μ m.

Adult *Chx10-Cre;Ptbp1^{lox/lox}* mice show only modest changes in gene expression in the retina

Despite the lack of any obvious defects in proliferation or cell fate specification in *Ptbp1*-deficient retinas, it nonetheless remained possible that this mutant might generate more subtle defects that are detectable using single-cell RNA-Seq analysis of whole retina. We therefore generated scRNA-Seq libraries from both P30 *Chx10-Cre;Ptbp1^{+/+}* and *Chx10-Cre;Ptbp1^{lox/lox}* mice. No changes in the identity or composition of any retinal cell type were observed (Fig. 5a-c). A modest number of genes in rod and cone photoreceptors, as well as bipolar neurons, showed statistically significant changes in gene expression (Fig. S4, Extended Data 2). A subset of phototransduction (*Pde6g*, *Guca1a*, *Pdc*, *Rcvrn*), photoreceptor-specific structural components (*Rom1*), and rod-specific transcription factors *Nrl* and *Nr2e3*, all showed modest but significant increases in expression in rods, while transcription factors such as *Otx2* and *Prdm1*, which are more prominently expressed in immature photoreceptors, showed significantly reduced expression. Cone photoreceptors showed increased expression of some phototransduction components (*Pde6c*, *Rcvrn*, *Guca1a*) and decreased expression of others (*Arr3*).

The largest number of differentially-expressed genes were observed in Müller glia (Fig. 5d). Several of the most strongly upregulated genes – notably including *Ptbp2*, *Fxyd6*, *Mt1*, and *Mt3* – were previously found to be upregulated by selective loss of function of *Ptbp1* in mature Müller glia in *Glast-CreER;Ptbp1^{lox/lox}* mice, with nearly a third of all significantly downregulated genes shared between both mutants (Hoang et al. 2022)(Fig. 5e, Extended Data 2). However, no significant changes in canonical markers of Müller glia such as *Sox9*, *Apoe*, *Glul*, or *Rlbp1* were observed (Fig. 5f).

Discussion

Previous studies have reported that *Ptbp1* acts in neural progenitors in the CNS as a master repressor of neuronal fate (Shibasaki et al. 2013; Hu et al. 2018). If this were the case, we would expect that *Ptbp1* loss of function in early development would rapidly lead to precocious cell cycle exit and reduction in the number of progenitor cells. As a result, we would expect to observe an overrepresentation of early-born cell types such as retinal ganglion cells and cone photoreceptors at the expense of late-born cell types such as bipolar interneurons and, in particular, Müller glia. However, we observe that while *Ptbp1* is robustly expressed in both retinal progenitors and mature glia, as in other CNS regions, loss of *Ptbp1* function in early-stage retinal progenitors does not induce any significant developmental defects. Despite efficient and widespread loss of *Ptbp1* expression from E14 onwards, we do not observe any changes in progenitor proliferation, retinal cell type composition, or expression of canonical Müller glia marker genes in mature retina. The changes in gene expression that are observed in adult *Ptbp1*-deficient retinas are quite modest, although increased levels of several photoreceptor-specific genes are observed. This is consistent with previous studies showing that *Ptbp1* knockdown enhances photoreceptor-specific splicing of a subset of alternatively spliced genes (Ling et al. 2020), and implies that this may indirectly increase expression of a subset of rod-specific genes.

At E16, however, homozygous *Ptbp1* mutants do show changes in gene expression consistent with previously reported functions. These retinas show increased expression of *Ptbp2*, the neuronally-enriched paralog of *Ptbp1*, along with increased expression of genes specific to late-stage progenitors and photoreceptor precursors. Furthermore, *Ptbp1* mutants exhibit altered RNA splicing and inclusion of a subset of exons specific to rod photoreceptor. While this is apparently consistent with reports that developmental loss of *Ptbp1* function promotes neurogenesis by

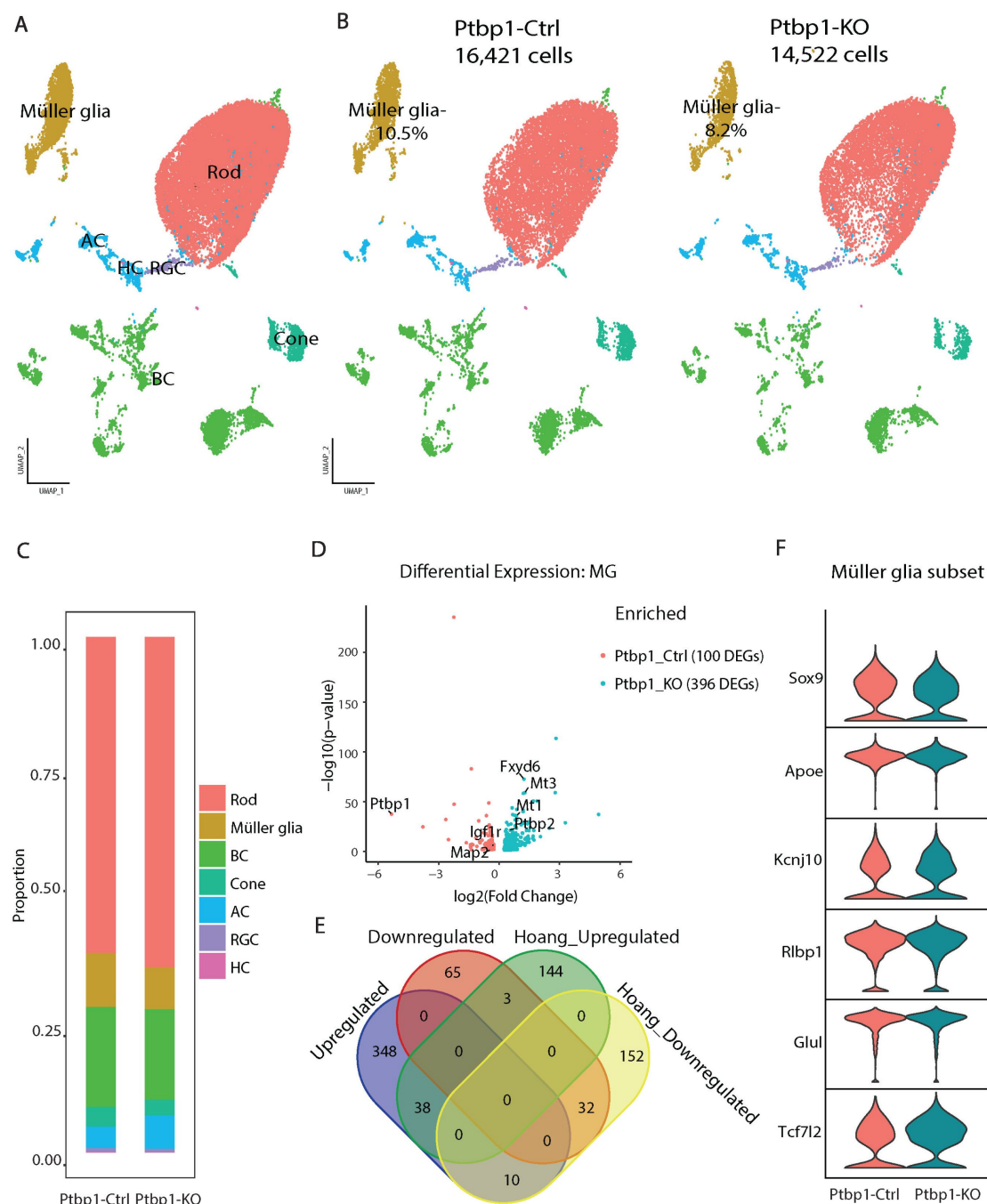


Figure 5.

scRNA-seq analysis reveals modest transcriptional changes in Müller glial gene expression in *Chx10-Cre;Ptbp1^{lox/lox}* mice.

(A) Uniform manifold approximation and projection (UMAP) plots showing cell types captured in the *Chx10-Cre;Ptbp1^{+/+}* (Ptbp1-Ctrl) and *Chx10-Cre;Ptbp1^{lox/lox}* (Ptbp1-KO) merged dataset. **(B)** UMAP plots showing cell types separated by genotype with the percentage of Müller glia in each group. **(C)** Barplot showing cell proportions for each cell type in both genotypes. **(D)** Volcano plot showing differentially expressed genes in Müller glia between Ptbp1-Ctrl and Ptbp1-KO samples. **(E)** Venn diagram comparison of upregulated and downregulated differentially expressed genes for Müller glia between this dataset and previously published data analyzing Müller glia-specific loss of function of *Ptbp1* (Hoang et al. 2022). **(F)** Violin plot showing no significant changes in the expression of canonical Müller glia markers in Ptbp1-KO sample compared to Ptbp1-Ctrl.

inducing a neuronal-like splicing pattern in neural progenitors (Hu et al. 2018 [DOI](#); Shibasaki et al. 2013 [DOI](#); Vuong et al. 2016 [DOI](#); Boutz et al. 2007 [DOI](#)), this does not result in changes in either progenitor proliferation or any change in retinal cell type composition in either the developing or adult retina. This argues against the idea that *Ptbp1* loss simply accelerates the timing of neurogenesis, as there is no evidence for increased photoreceptor production or altered proliferation in embryonic or neonatal retina. An alternative interpretation would be that *Ptbp1* loss of function leads to the precocious formation of ectopic neurons which rapidly undergo cell death. Further studies will be needed to resolve this question.

Following closely behind studies that fail to show any role for *Ptbp1* in regulating glia-to-neuron reprogramming in adult CNS (L.-L. Wang et al. 2021 [DOI](#); Hoang et al. 2023 [DOI](#), 2022 [DOI](#); Chen et al. 2022 [DOI](#)), these findings call into question previous models that place *Ptbp1* in a central position in controlling developmental neurogenesis. The reason for this discrepancy is not entirely clear. Previous studies using knockdown may have been complicated by off-target effects (Jackson et al. 2003 [DOI](#)), and conditions for *in vitro* analysis may have accurately replicated conditions in the native CNS. Alternatively, upregulation of the widely expressed paralog *Ptbp2* may have compensated for developmental defects resulting from *Ptbp1* loss of function. Generation of conditional double mutant mice will be necessary to definitively exclude this possibility. Regardless, this study demonstrates that *Ptbp1* is dispensable for the process of retinal neurogenesis and cell fate specification, and we anticipate that this will likely hold for other CNS regions.

Methods

Mice

Mice were raised and housed in a climate-controlled pathogen-free facility on a 14/10 h light/dark cycle. Mice used in this study were *Chx10-Cre* (JAX #005105) *Ptbp1*^{lox/lox} mice carrying loxP sites that flank the promoter and 1st exon of *Ptbp1* were generated as described previously (Shibayama et al., 2009). *Chx10-Cre;Ptbp1*^{lox/lox} mice were generated by crossing *Chx10-Cre* with conditional *Ptbp1*^{lox/lox} mice. Maintenance and experimental procedures performed on mice were in accordance with the protocol approved by the Institutional Animal Care and Use Committee (IACUC) at the Johns Hopkins School of Medicine under protocol number MO22M22.

EdU treatment

Chx10-Cre control and *Chx10-Cre;Ptbp1*^{lox/lox} neonatal mice were administered EdU (Abcam, #ab146186) via subcutaneous injection at P0 [10–20 µl of PBS (5 mg/ml)]. Retinas were collected two hours post EdU injection for analysis.

Immunohistochemistry and imaging

Collection and immunohistochemical analysis of retinas were performed as described previously (Hoang et al., 2020 [DOI](#)). Briefly, mouse eye globes were fixed in 4% paraformaldehyde (ElectronMicroscopySciences, #15710) for 1 h at 4°C. Retinas were dissected in 1x PBS and incubated in 30% sucrose overnight at 4°C.

Retinas were then embedded in OCT (VWR, #95057-838), cryosectioned at 16 µm thickness, and stored at –20°C. Sections were dried for 30 min in a 37°C incubator and washed 3 x 5 min with 0.1% TritonX-100 in PBS (PBST). EdU labeling was performed by using Click-iT EdU kit (ThermoFisher, #C10340) following the manufacturer's instructions. Sections were then incubated in 10% Horse Serum (ThermoFisher, #26050070), 0.4% TritonX-100 in 1x PBS (blocking buffer) for 2 h at room temperature (RT) and then incubated with primary antibodies in the blocking buffer overnight at 4°C. Primary antibodies used in this study were: rabbit anti-*Ptbp1* (1:200, Invitrogen,

#PA3-81297), rabbit anti-Ptbp1 (1:200, ABclonal, #A6107), rabbit anti-RBPMS (1:200, Proteintech, #151871-AP), rabbit anti-cone arrestin (1:200, Millipore Sigma, #AB15282), goat anti-OTX2 (1:200, R&D Systems, #AF1979), mouse anti-HuC/D (1:200, ThermoFisher Scientific, #A21271), mouse anti-NeuN (1:200, SigmaAldrich, #MAB377), rabbit anti-SOX9 (1:200, SigmaAldrich, #AB5535) and chicken anti-GFP (1:400, ThermoFisher Scientific, #A10262). An equal mixture of anti-HuC/D and NeuN antibodies was used to detect amacrine and ganglion cells.

Sections were washed 3 x 5 min with PBST to remove excess primary antibodies and were incubated in secondary antibodies in blocking buffer for 2 h at RT. Sections were then counterstained with DAPI in PBST, washed 3 x 5 min in PBST and mounted with ProLong Gold Antifade Mountant (Invitrogen, #P36935) under coverslips (VWR, #48404-453), air-dried, and stored at 4°C. Fluorescent images were captured using a Zeiss LSM 700 confocal microscope.

Cell quantification and statistical analysis

Ptbp1-positive, Sox9-positive, HuC/D/NeuN-positive, Otx2-positive, Rbpms-positive, and ArrestinC-positive cells were counted in total from 300µm sections per retina. For cell proliferation quantification, EdU-positive cells were counted from 300µm sections per retina. Each data point in the bar graphs was calculated from an individual retina. All cell quantification data were graphed and analyzed using GraphPad Prism 10. Multiple unpaired *t* tests were used for analysis between samples of multiple groups. In all tests, values of *p* < 0.05 were considered to indicate significance.

Retinal cell dissociation

Briefly, one female mouse per genotype was euthanized by CO₂. Retinas were dissected in fresh ice-cold PBS and retinal cells were dissociated using Papain Dissociation System (LK003150, Worthington). Dissociated cells were resuspended in ice-cold HBAG buffer containing 9.8mL HibernateA (BrainBits, #HALF500), 200mL B27 (ThermoFisher, #17504044), 20mL GlutaMAX (ThermoFisher, #35050061) and 0.5U/mL RNase inhibitor. Cells were filtered through a 50mm filter. Cell count and viability were determined by using 0.4% Trypan blue.

Bulk RNA-sequencing library preparation

Retinas were dissected from E16 mice embryos and stored at -80°C. Frozen retinal tissue was pooled into 7 samples by genotype: wildtype (WT (6 pooled retinas)); heterozygous *Ptbp1* mutant *Chx10-Cre;Ptbp1^{lox/+}* (Het1 (5 pooled retinas), Het2 (5 pooled retinas), Het3 (6 pooled retinas)); and homozygous *Ptbp1* mutant (KO1 (4 pooled retinas), KO2 (4 pooled retinas), and KO3 (6 pooled retinas)). RNA was extracted using the RNeasy Micro Plus kit (Qiagen). Illumina Stranded mRNA Prep Ligation kit was used for bulk RNA library preparation. Libraries were sequenced on the Illumina NovaSeq X.

Single-cell RNA-sequencing (scRNA-seq) library preparation

scRNA-seq were prepared on dissociated retinal cells using the 10X Genomics Chromium Single Cell 3' Reagents Kit v3.1 (10X Genomics, Pleasanton, CA). Libraries were constructed following the manufacturer's instructions and were sequenced using the Illumina NovaSeq 6000. The sequencing data were aligned to the mm10 reference genome and a cell-by-gene matrix was generated using the Cell Ranger 7.0.1 pipeline (10X Genomics).

ScRNA-seq data analysis

ScRNA-seq of control *Chx10-Cre;Ptbp1^{+/+}* and homozygous mutant *Chx10-Cre;Ptbp1^{lox/lox}* mice were analyzed using the Seurat package (Hao et al. 2021 [\[1\]](#)). Briefly, cells were filtered to retain those with more than 200 and fewer than 5,000 detected genes, more than 700 and fewer than 10,000 total transcripts, and less than 20% mitochondrial gene content. The data were normalized using the default log-normalization parameters, the top 4000 highly variable features were

identified, and data were scaled while regressing out mitochondrial gene content. Principal component analysis (PCA) was then performed on the variable genes followed by louvain clustering and UMAP visualization using the top 12 principal components. Expression of key marker genes within each cell cluster were used to confirm the appropriate assignment of cell types. Differentially expressed genes between groups were determined using the Wilcoxon rank sum test. The clusterProfiler R package was used to perform gene ontology enrichment analysis of biological processes found in the differentially expressed gene lists (Yu et al. 2012 [↗](#)).

Bulk RNA-seq analysis

Reads were processed using the rMATS-turbo pipeline (Y. Wang et al. 2024 [↗](#)), which performs alignment with STAR and custom splice junction counts. Alignments were generated using the GRCm38 mouse genome (Release M10) with GENCODE annotation. Gene-level counts were then extracted from the resulting BAM files using featureCounts, and differential gene expression was analyzed with the DESeq2 package in R (Liao, Smyth, and Shi 2014 [↗](#); Love, Huber, and Anders 2014 [↗](#)). Comparisons between *Ptbp1* heterozygous (Het) and knockout (KO) samples identified differentially expressed genes based on an absolute log2 fold change greater than 0.263 (representing at least a 20% change in expression) and an adjusted p-value below 0.05. We incorporated publicly available bulk RNA-seq datasets for comparative analysis, including embryonic day 16 retina (E16: GSM2720095, GSM2720096), postnatal day 28 retina (P28: GSM2720111, GSM2720112), Thy1-positive neurons (GSM2392791), astrocytes (GSM1269903, GSM1269904), microglia (GSM1269913, GSM1269914), hair cells (GSM1602228, GSM1602229), and heart tissue (GSM1223635) (Brooks et al. 2019 [↗](#); Yang et al. 2017 [↗](#); Y. Zhang et al. 2014 [↗](#); Cai et al. 2015 [↗](#); Giudice et al. 2014 [↗](#)). Hair cells and heart tissue samples were chosen as examples of high and low *Ptbp1* expression, respectively, based on data from ASCOT (Ling et al. 2020 [↗](#)). We generated a Pearson correlation matrix to assess overall sample similarity. Splicing differences were evaluated using filtered rMATS output, applying the following thresholds: read coverage ≥ 15 , minimum PSI = 0.05, maximum PSI = 0.95, significant FDR (sigFDR) ≤ 0.05 , $|\Delta\text{PSI}| \geq 0.10$, background FDR (bgFDR) ≤ 0.5 , and background within-group $\Delta\text{PSI} \leq 0.2$. For the E16 and P28 retina datasets, we used more relaxed filters (read coverage ≥ 0 , PSI between 0.01 and 0.99) to ensure inclusion of lower-abundance events. Splicing events specific to rods and various neuronal subtypes were queried from the ASCOT supplementary files and visually inspected using the UCSC Genome Browser to assess their presence in bulk RNA-seq datasets. This analysis followed the approach described in previous studies (Carmen-Orozco et al. 2024; Ling et al. 2020 [↗](#)).

Data availability

All raw RNA-Seq and scRNA-Seq data are publicly available through the Gene Expression Omnibus under accession numbers GSE300588 and GSE300607. Analysis scripts used in this study are accessible at: https://github.com/csanti88/ptbp1_2025 [↗](#). Bulk RNA-seq and other tissue track plots visualizations are available at: https://genome.ucsc.edu/s/roggercarmen/Ptbp1KO_Retina [↗](#)

Figure supplements

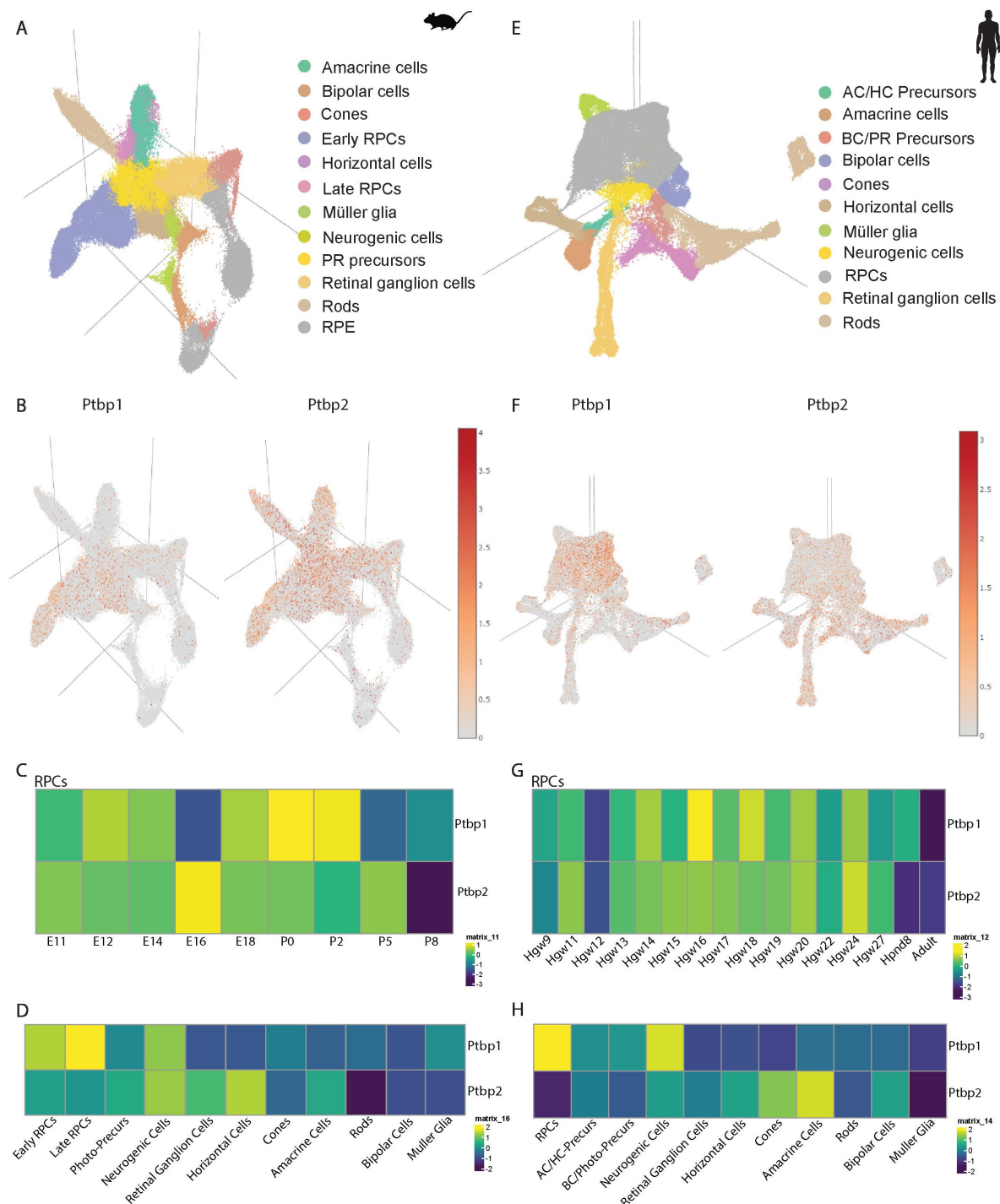


Figure S1.

Cellular expression patterns of *Ptbp1* and *Ptbp2* during mouse and human retinal development.

(A) 3D UMAP plot showing cell types from combined timepoints (E14-Adult) of mouse retinal development utilizing publicly available data from (Clark et al. 2019). (B) Gene plots for *Ptbp1* and *Ptbp2* expression during mouse retinal development. (C) Heatmaps of *Ptbp1* and *Ptbp2* expression patterns in RPCs at varying time points during mouse retinal development. (D) Heatmaps of *Ptbp1* and *Ptbp2* expression patterns in different mouse retinal cell types. (E) 3D UMAP plot showing cell types from combined timepoints (Hgw9-Adult) of human retinal development utilizing publicly available data from (Lu et al. 2020). (F) Gene plots for *Ptbp1* and *Ptbp2* expression during human retinal development. (G) Heatmaps of *Ptbp1* and *Ptbp2* expression patterns in RPCs at varying time points during human retinal development. (H) Heatmaps of *Ptbp1* and *Ptbp2* expression patterns in different human retinal cell types.

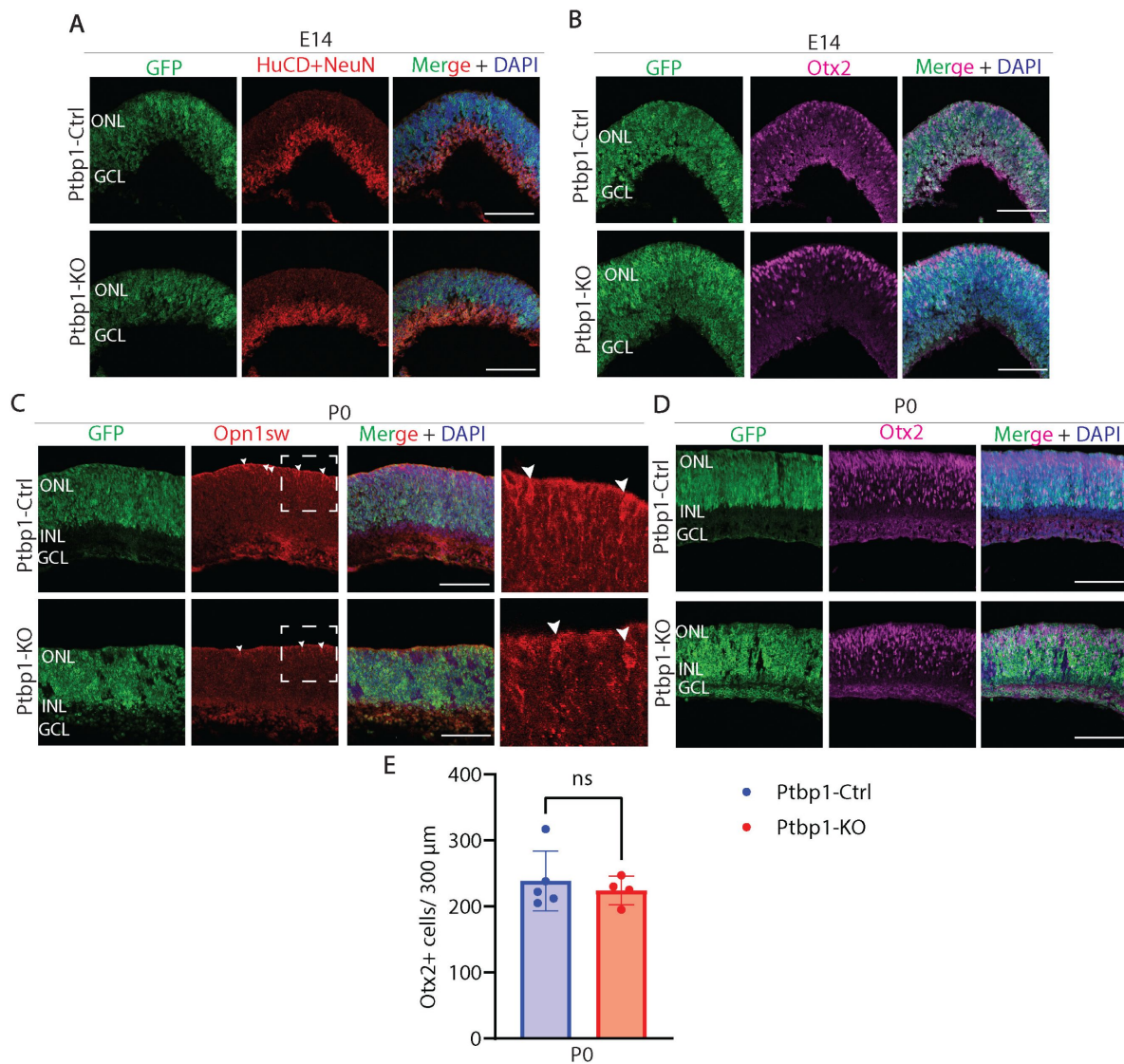


Figure S2.

Immunostaining does not detect changes in cell composition in E14 or P0 *Ptbp1*-deficient retina.

(A) Representative immunostaining for, HuC/D+NeuN and **(B)** Otx2 in *Ptbp1*-Ctrl and *Ptbp1*-KO mice at E14. **(C)** Representative immunostaining for Opn1sw and **(D)** Otx2 in *Ptbp1*-Ctrl and *Ptbp1*-KO mice at P0. **(E)** Quantification of Otx2-positive cells in *Ptbp1*-Ctrl and *Ptbp1*-KO retinas at P0 ($n \geq 4$ /genotype). Significance was determined using an unpaired t test: ns= $p > 0.05$. Each data point was calculated from an individual retina. ONL, outer nuclear layer; INL, inner nuclear layer; GCL, ganglion cell layer. Scale bar= 100 μm .

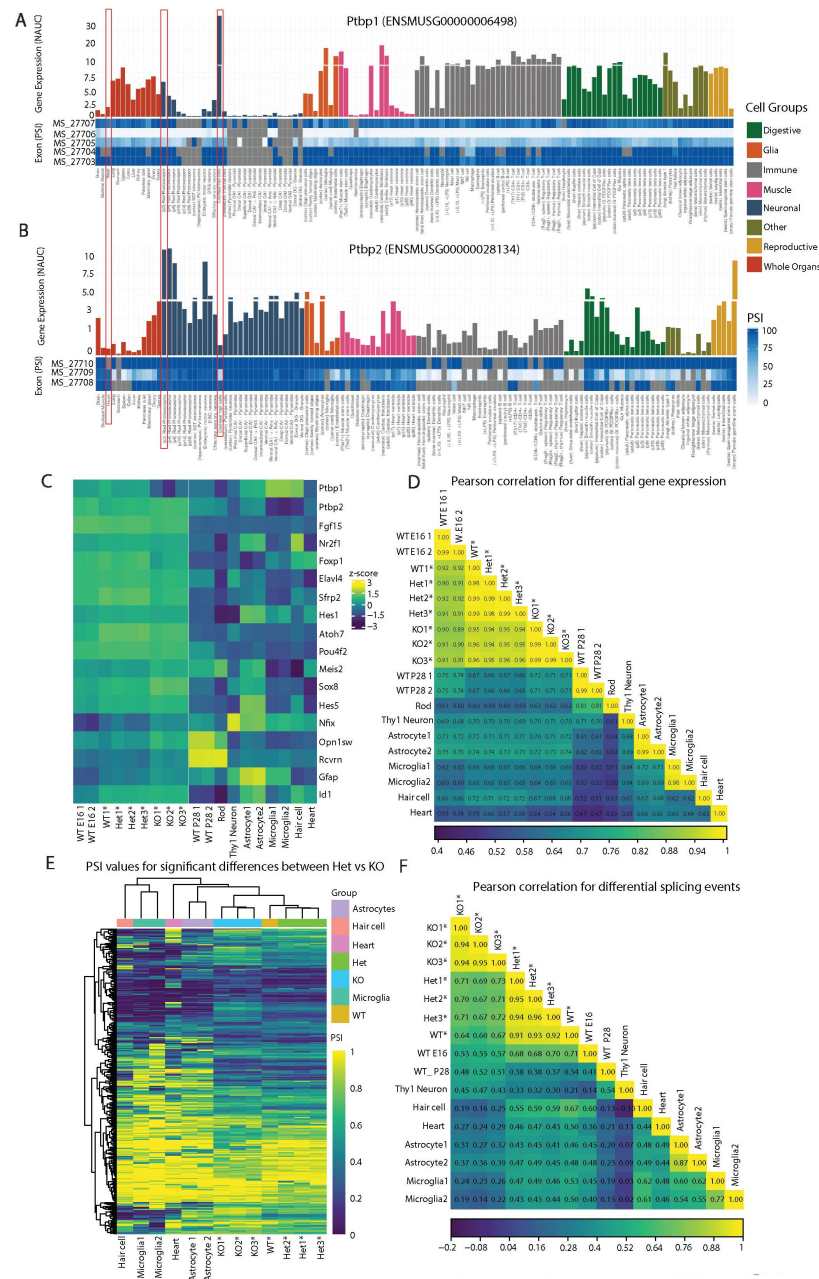


Figure S3.

BulkRNA-seq analysis reveals altered gene expression and splicing patterns in E16 *Chx10-Cre;Ptpb1^{lox/lox}* mice retina.

(A) Bar charts showing *Ptpb1* and (B) *Ptpb2* expression in various cell types and tissues using the ASCOT browser (Ling et al. 2020). Highlighted samples are rods photoreceptors at P2, heart tissue showing low *Ptpb1* expression and cochlear hair cells showing high *Ptpb1* expression. (C) Heatmap of select differentially expressed genes between individual replicates samples for wildtype (WT), heterozygous *Chx10-Cre;Ptpb1^{lox/+}* (Het1, Het2, Het3), and homozygous *Chx10-Cre;Ptpb1^{lox/lox}* (KO1, KO2, KO3) *Ptpb1* mutant retinal samples analyzed in this study, along with publicly available bulk RNA-seq datasets including embryonic day 16 retina, postnatal day 28 retina, Thy1-positive neurons, astrocytes, microglia, hair cells and heart tissue. (D) Pearson correlation plot for identified differentially expressed genes between Het and KO samples, plotted across all samples including publicly available datasets used. (E) Heatmap showing PSI values and (F) pearson correlation plot for splicing events identified as differentially spliced between Het and KO samples, plotted across all samples including publicly available datasets.

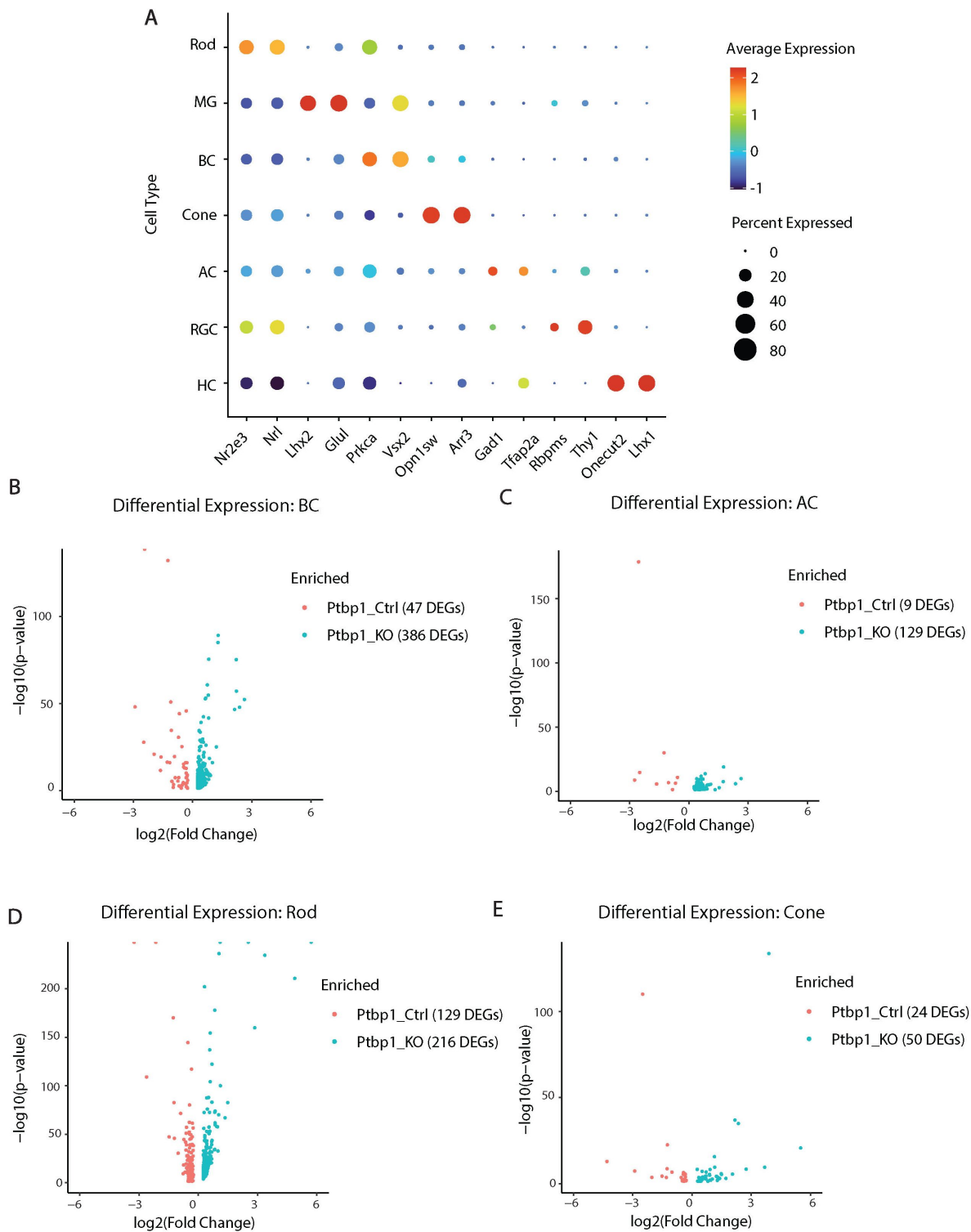


Figure S4.

ScRNA-seq analysis identifies differentially expressed genes in major adult retinal cell types following developmental loss of function of *Ptbp1*.

(A) Dot Plot of cell type-specific markers used to identify clusters for each major retinal cell type. (B) Volcano plot showing differentially expressed genes for bipolar cells (BC), (C) amacrine cells (AC), (D) rod photoreceptors and (E) cone photoreceptors between *Ptbp1*-Ctrl and *Ptbp1*-KO samples.

Acknowledgements

This work was supported by NIH grant R01EY036173 to S.B.

Additional files

Extended Dataset 1. Bulk RNA-Seq analysis. Table of 2075 differentially expressed genes between heterozygous *Chx10-Cre;Ptp1^{lox/l+}* and homozygous *Chx10-Cre;Ptp1^{lox/lox}* mutants identified by bulk RNA-Seq at E16.5, table of 864 differential splicing changes, and table of adult neuronal and rod enriched splicing events that overlap with the differential splicing events in the bulk RNA-Seq dataset. [↗](#)

Extended Dataset 2. scRNA-Seq analysis. Proportions of cell types in the *Ptp1-Ctrl* and *Ptp1-KO* scRNA-Seq datasets. Table of differentially expressed genes between *Chx10-Cre;Ptp1^{+/+}* control and homozygous *Chx10-Cre;Ptp1^{lox/lox}* mutants identified by scRNA-Seq in adult retina for Müller glia (496), bipolar cells (433), amacrine cells (138), rod photoreceptor (345) and cone photoreceptor cells (74) with adjusted *p*-value of 0.05. Positive *avg_log2FC* scores represent upregulated genes within the *Ptp1-KO* sample. [↗](#)

Additional information

Funding

National Eye Institute (R01EY036173)

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Reviewer #1 (Public review):

Summary:

The researchers sought to determine whether Ptbp1, an RNA-binding protein formerly thought to be a master regulator of neuronal differentiation, is required for retinal neurogenesis and cell fate specification. They used a conditional knockout mouse line to remove Ptbp1 in retinal progenitors and analyzed the results using bulk RNA-seq, single-cell RNA-seq, immunohistochemistry, and EdU labeling. Their findings show that Ptbp1 deletion has no effect on retinal development, since no defects were found in retinal lamination, progenitor proliferation, or cell type composition. Although bulk RNA-seq indicated changes in RNA splicing and increased expression of late-stage progenitor and photoreceptor genes in the mutants, and single-cell RNA-seq detected relatively minor transcriptional shifts in Müller glia, the overall phenotypic impact was low. As a result, the authors conclude that Ptbp1 is not required for retinal neurogenesis and development, thus contradicting prior statements about its important role as a master regulator of neurogenesis. They argue for a reassessment of this stated role. While the findings are strong in the setting of the retina, the larger implications for other areas of the CNS require more investigation. Furthermore, questions about potential reimbursement from Ptbp2 warrant further research.

Strengths:

This study calls into doubt the commonly held belief that Ptbp1 is a critical regulator of neurogenesis in the CNS, particularly in retinal development. The adoption of a conditional knockout mouse model provides a reliable way for eliminating Ptbp1 in retinal progenitors while avoiding the off-target effects often reported in RNAi experiments. The combination of bulk RNA-seq, scRNA-seq, and immunohistochemistry enables a thorough examination of molecular and cellular alterations at both embryonic and postnatal stages, which strengthens the study's findings. Furthermore, using publicly available RNA-Seq datasets for comparison improves the investigation of splicing and expression across tissues and cell types. The work is well-organized, with informative figure legends and supplemental data that clearly show no substantial phenotypic changes in retinal lamination, proliferation, or cell destiny, despite identified transcriptional and splicing modifications.

Weaknesses:

The retina-specific method raises questions regarding whether Ptbp1 is required in other CNS locations where its neurogenic roles were first proposed. The claim that Ptbp1 is "fully dispensable" for retinal development may be toned down, given the transcriptional and splicing modifications identified. The possibility of subtle or transitory impacts, such as ectopic neuron development followed by cell death, is postulated, but not completely investigated. Furthermore, as the authors point out, the compensating potential of increased Ptbp2 warrants additional exploration. Although the study performs well in transcriptome and histological analyses, it lacks functional assessments (such as electrophysiological or behavioral testing) to determine if small changes in splicing or gene expression affect retinal

function. While 864 splicing events have been found, the functional significance of these alterations, notably the 7% that are neuronal-enriched and the 35% that are rod-specific, has not been thoroughly investigated. The manuscript might be improved by describing how these splicing changes affect retinal development or function.

<https://doi.org/10.7554/eLife.108331.1.sa1>

Reviewer #2 (Public review):

Summary:

Ptbp1 has been proposed as a key regulator of neuronal fate through its role in repressing neurogenesis. In this study, the authors conditionally inactivated Ptbp1 in mouse retinal progenitor cells using the Chx10-Cre line. While RNA-seq analysis at E16 revealed some changes in gene expression, there were no significant alterations in retinal cell type composition, and only modest transcriptional changes in the mature retina, as assessed by immunofluorescence and scRNAseq. Based on these findings, the authors conclude that Ptbp1 is not essential for cell fate determination during retinal development.

Strengths:

Despite some effects of Ptbp1 inactivation (initiated around E11.5 with the onset of Chx10-Cre activity) on gene expression and splicing, the data convincingly demonstrate that retinal cell type composition remains largely unaffected. This study is highly significant since it challenges the prevailing view of Ptbp1 as a central repressor of neurogenesis and highlights the need to further investigate, or re-evaluate, its role in other model systems and regions of the CNS.

Weaknesses:

A limitation of the study is the use of the Chx10-Cre driver, which initiates recombination around E11. This timing does not permit assessment of Ptbp1 function during the earliest phases of retinal development, if expressed at that time.

<https://doi.org/10.7554/eLife.108331.1.sa0>